Crime Scene Object Detection for Forensic Investigations Using Faster R-CNN and YOLOv5 Models

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***Abstract*—More and more complex and numerous are the forensic findings required while investigating the crime scene, it is important to address the need for technological enhancements for the crime scene analysis. This objective detection models are important in helping alleviate the numerous errors that are involved when a human is tasked with the responsibility of identifying and categorizing objects within a crime scene, as it faster the process. In this research, Faster R-CNN and YOLOv5 deep learning models are used to detect the objects in crime scenes. Faster R-CNN which offers accuracy in object detection is used while YOLOv5 a real-time object detection framework improves the speed of detection. The models were trained and tested on a dataset which contains images of crime scene and the related objects include weapon, evidence mark and personal effect. The efficiency of the developed models was assessed by comparing the results on the mAP, detection speed, and false positive ratios. As such, the experimental results show that YOLOv5 is faster than Faster R-CNN for real-time applications, whereas Faster R-CNN is more accurate for higher detection rate-based applications. These models are complementary in their operation, the study suggests an integration of these models to improve efficiency for the forensic process. The study shows that by incorporating data object detection into contemporary forensic investigation processes powered by AI, the forensic science will significantly improve its ability to analyze evidence and solve crimes.**

Keywords— Object Detection, Crime Scene Analysis, YOLOv5, Faster R-CNN, Forensic Investigation

I. INTRODUCTION

Forensic science as a discipline has always depended on the classification and examination of samples originating at scenes of the crimes. With the passage of time, use of digital tools in the process of investigative action helps to get a more accurate examination of the crime scene. Among these achievements one can identify the use of deep learning methodologies, especially when it comes to methods for object detection[1]. Object detection, a task of detecting and categorizing objects in image or video sequences, is now a key component in crime scene analysis, where cataloguing physical evidence can be tedious and error-prone. AI methods like Faster R-CNN and YOLOv5 in particular have the potential to increase the efficiency, effectiveness and credibility of crime scene analysis and for these reasons can be of significant use for forensic analysis. There are two of the most modern approaches for object detection nowadays: Faster R-CNN (Region-based Convolutional Neural Network) and YOLOv5 (You Only Look Once version 5). Region-based procedural approach include the faster R-CNN that uses both convolutional neural networks and the region proposal network to enhance the detection accuracy. On the other hand, YOLOv5 is a real-time detection model while its emphasis is on speed and efficacy it is highly valuable in a scenario where detection is more urgently needed. While applications of SV-ICP-DA and DGI have been shown prominent success in numerous contexts, the possibility of their implementation in forensic science has not been investigated broadly[2].

Within the framework of crime scene analysis, object detection models are required to analyze a vast potential list of include weapons, clothes, vehicles, personal effects, and markers[1]. This shall may help the investigators in the reconstruction of events and general surrounding occasions of a crime thus improving on accuracy. Furthermore, an intelligent system that can quickly search for these objects can be useful to the investigators and relieve the necessity of the identification process from them. Thus, lighting that might be present at a crime scene, or the occlusion, and clutter, must not influence the chosen model in a negative way but, on the contrary, the model has to be stable and reliable and provide good enough results[3]. Much of the prior work has been directed toward deep learning models in the context of object detection in security, health care, and self-driving car applications. However, to the best of the researcher's knowledge, research anchoring these models to crime scene object detection is lacking. There are only a few works, and the majority of them is aimed at identifying objects in general without the specific characteristics that investigations in the field of forensics necessitate. For instance, object detection in forensic analysis has several oddities that must be solved, for example, similar items such as guns and knives have to be differentiated, the images to work with can be low-resolution,

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captured by surveillance cameras or crime scene photos[4]. This research seeks to address this research question by examining and comparing Faster R-CNN and YOLOv5 for crime scene object detection. Therefore, in comparing reliable variable accuracy rate, speedy and adaptable models to these models, the study aims at identifying the most appropriate method for real life forensic use. The goal is primarily to achieve increased performance in forensic investigations through accelerated processing of evidence items while preserving a high degree of accuracy in object detection[5].

To this end, the study employs a dataset of crime scenes images, with annotated objects that are commonly encountered in forensics. This data set will be used to train and evaluate both the models using mAP, speed of detection and False Positive rates[6]. Thus, this research intends to discuss various models comprehensively on the ways in which they may be used to analyze crime scenes in order to identify unique features that may make one model preferable to another. In the next sections, the process of training and validation of the models is described, and then the obtained outcomes are reviewed. The study also considers the possibility of integrating the models or applying them in different settings: for example, with real-time video or in a crime scene investigation. Thus, the research aims to advance the current scholarly conversation on the application of AI and Machine Learning as an added component to the forensic science practicing today by law enforcement agencies and forensic experts[7].

II. Literature Review

Due to the increased interest of the use of deep learning in object detection, different techniques have been formulated and used for different purposes. However, two of the most researched and applied techniques are listed below, Faster R-CNN and YOLOv5[8]. These models have found applications in various domains of life including, self-driving cars, medical imaging, surveillance, etc., but its use in forensic sciences is fairly new territory. Thus, a literature review shows that there is significant interest in applying deep learning models for crime scene analysis but there is much work to be done in order to fine-tune models for forensics. Another kind of object detection technique was Faster R-CNN proposed by Ren and his colleagues in 2015. The pre-nets are combined with a Region Proposal Network (RPN) and a Fast R-CNN detection network thus enhancing the speed and performance of object detection. After that, people use Faster R-CNN for high-precision object detection tasks, such as face recognition, medical image analysis, and surveillance[9]. The performance of Faster R-CNN in detecting small objects and objects that are surrounded by other prominent objects, which is the case in crime scene investigations, has also been investigated in many other related researches. In the forensic context, morphing of images and objects, low resolution, or presence of excessive noise and occlusions significantly hampers the task; it is highly important to detect objects such as weapons or evidence markers in the backgrounds or at the corners of the images. Lin et al. (2017) and Liu et al. (2020) demonstrate Faster R-CNN's flexibility to detect objects in difficult situations, which is the reason it should be utilized for crime scene object identification[10].

Nevertheless, setting aside the high accuracy that is provided by Faster R-CNN, it turned out to be slower than other processes, and that can be critical for further analyses in real-time applications. It is here that models such as YOLO (You Only Look Once) have proved more useful. YOLO proposed by Redmon et al. (2016) disrupted the method to perform true-time object detection solely as a single regression problem, which is much faster compared to region-based like Faster R-CNN. , the latest family of YOLO called YOLOv5 is shines in terms of speed without being inferior and sometimes surpassing the accuracy of other models[11]. Based on the findings by Glenn Jocher et al. (2020), YOLOv5 scales the speed benefits of earlier YOLO versions alongside the accuracy enhancements to its predecessors making YOLOv5 suitable for real time use. A rich advantage of the proposed approach is image processing capability, especially in applications such as video surveillance or crime scene video streaming, where timely object detection can play an important role in the process. Recently deep learning based object detection models are being adopted in forensic science particularly in automating the process of detection and classification of objects and-items in crime scene[12]. As for the early articles like Choi et al. (2017) it is possible to speak about usage of the traditional Computer Vision approaches, including the edge detection and the feature matching, to define the objects in the crime scene images. Althougn these methods were useful for their purposes, none of them features the precision and flexibility of today's deep learning methods[13]. Transition from conventional learning to deep learning started with the use of convolutional neural networks which was accurate in the determination of objects in a complicated manner. Lately, there are works that investigate the usage of Faster R-CNN and YOLO models in the forensic context. For example, Purohit et al. (2019) used Faster R-CNN as a method of identifying and categorizing the items in crime scene photos with moderate levels of success but that open the caveat that further improvement of the model is needed to counteract problems such as occlusion or differing image quality[14].

For example, one of the main difficulties encountered when trying to use deep learning models in the context of crime scene object detection, comes from differences in image quality and from the objects' occlusion, which are common in forensic images. Crime scene photographs are often captured in suboptimal circumstances; photographs may be taken at low light or with objects partially occluded by others[15]. These challenges make it difficult for traditional object detection models of classifying objects in real life scenarios. These difficulties, however, form the subject of the most recent studies attempting to surmount them. Zhang, Xia, et al. (2020) introduced a novel approach to the original Faster R-CNN model that introduces attentiveness to the occluded objects that are present in the crime scene images. In the same way, Wei et al. (2021) proposed a data augmentation technique to enhance the ability of YOLOv5 to handle different conditions in images. These progression shows that deep learning models have the capability to develop to meet the complex difficulties in forensic images[16]. The majority of the prior research has concentrated on deploying a single model, such as Faster R-CNN or YOLOv5, etc., in the object detection process; however, recent research has started to investigate the use of multiple models for enhanced object detection. For example, Zhang et al. (2022) suggested integrating Faster R-CNN with YOLOv4 because these two approaches are known to work best when combined since each ostensively has the other's shortcomings[17]. This approach of having a mixture of the two models might be useful for

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crime scene object detection, where some of the objects might need very high precision of Faster R-CNN and others might not need the elaborate ability of YOLOv5 but can be very fast. The possibility for developing the synergism of these models in a manner that will allow them to be rather accurate and rather fast is one of the current trends in the development of AI for forensic analysis[18]. Therefore, even though conventional object detection in a number of scenarios have demonstrated the effectiveness of deep learning models, the forensic application of this tool is still quite nascent. That is why further development of approaches such as Faster R-CNN and YOLOv5 has proved high productivity both in terms of accuracy and time while certain issues concerning image quality and occlusion of the object still persist. These models have been described in a growing number of publications as having the ability to dramatically transform crime scene analysis, although the feasibility of these models for forensic purpose requires further study in order to fine-tune these models to work in the forensic environment. For future research, these models ought to be tested under conditions similar to real crime scenes, further discussions about the benefits of combining B-Spline models with other kinds of models should be conducted, and more experiments ought to be carried out to examine how the implementation of these models in the professional work of the forensic experts affects their efficiency.

III. Proposed Methodology

This research work put forward a recommendation strategy for the assessment and the comparative analysis of two state-of-art object detection techniques: Faster R-CNN and YOLOv5 to conduct the crime scene object detection. Its accuracy is that the methodology is intended to establish the rate of effectiveness as well as the efficiency of these models in analyzing and categorizing these objects in crime scene images.

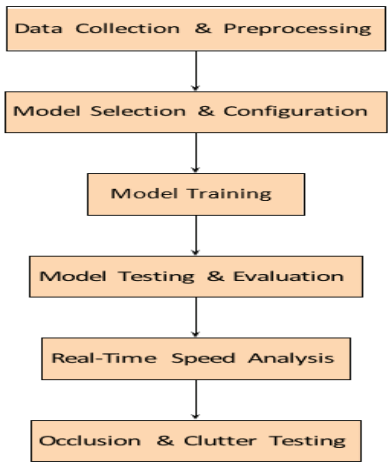

Fig. 1. Proposed Architecture for Crime Scene Object Detection Using Faster R-CNN and YOLOv5 Models

This method entails different process such as data gathering, building, evaluating the model, and comparing the outcome. The proposed methodology guarantees that both models are evaluated under optimal circumstance in order to compare their for and against and get the best result for forensic endeavors.

A. Data Collection and Preprocessing

The first stage of the methodology is to acquire suitable set of crime scene images with labeled objects that usually appears in the crime scene such as weapon, belongings, evidence tags among them and others as may be required. The dataset should also include variety of lighting conditions, occlusion of the object under investigation, image resolution and background clutter in order to replicate (as close as possible) the real-world situations that a forensic investigator encounters. The datasets can be derived from existing datasets as the Crime Scene Dataset (CSD) or other similar datasets or a new dataset can be used which can be collected by working with professional forensic intelligence. Resizing images and other similar manipulations such as normalization of pixels by applying a standard size and mean pixel value across all images, data augmentation techniques such as rotation, flipping and color jugeling will be applied to enhance the model's robustness and reduce over fitting.

B. Model Selection and Configuration

For this work, Faster R-CNN and YOLOv5 models will be adopted because of its prior credibility in carrying out object detection tasks. For forensic object detection due to their nature that requires high accuracy and inconsistency, particularly where the objects of interest are likely to be small and intricate-like weapons or markers-Faster R-CNN will be tuned relative to the precision that a particular object responsibility network can offer as it includes region of proposal network. Unlike YOLO, which has been trained with version 3, YOLOv5 will be set to take advantage of its high rate of processing to be used in real-time crime scenes analysis. Both models will be set to a Cohen's general object detection weights obtained from a general object detection set such as COCO and then retrained on the crime scene set.

C. Model Training

Practical training of the models will come in a form of appropriate deep learning framework like TensorFlow or PyTorch in which two main models will be implemented and trained: Faster R-CNN as well as YOLOv5. The models will be trained on a GPU-equipped system in order to accelerate the training. The landmarks of the different objects in the crime scene images will be recognized during training, with the help of minimizing the loss function that points at the difference between the predicted and the real object's bounding box and its class. It will then be fine-tuned depending on initial results that a learning rate, batch size, and number of epochs shall be fine-tuned before implementing it to the models. Moreover, cross-validation will be applied to control the models' performance during training and avoid a overfitting issue.

D. Model Testing and Evaluation

As described above, after training both models will be evaluated on another set of crime scene images which were not used during the training phase. The chosen metrics will be mean average precision (mAP), intersection over union (IoU) and precision-recall curves, where the former two is going to be used to measure the accuracy of object detection and its precision to the ground truth, and the latter is going to analyze the models' performance by its precision and recall. The Precision Recall curves will give cross sectional view of the precision and recall measures for varying detection thresholds. I argue that the results will illustrate which model is superior

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in the process of accurate object detection, while minimizing false positives.

E. Speed and Real-Time Analysis

Besides accuracy, the time taken to perform the object detection is effectively important when it comes to forensic applications. The real time detection option in YOLOv5 will be particularly tested focusing on images that are crime scenes. Two modes of inference time will be calculated: Often measured in milliseconds, the inference time refers to the time taken by a model to detect objects in a single image. This will be done by feeding the same test image's set to the two models and keep tab on the amount of time it will take to detect and categorize the objects we are to detect. A compromise between speed and accuracy will be discussed, especially where rapid detection is needed when the site is being used for crime scene analysis. These findings will therefore be used to assess how applicable the various models are in forensic scenarios.

F. Handling Object Occlusion and Clutter

Some things are partly occluded by other objects or are within cluttered scenes which makes detection difficult in images related to crime scenes. To this end, both models will be specifically tested on occluded objects and cluttered background. Partially occluded or overlapping objects of interest will be present in the test dataset. The effectiveness of the models shall be evaluated by the models' accuracy of identifying and categorizing these partially occluded or hidden objects. Moreover, data augmentation techniques will be introduced during training to covers the occlusion and clutter problem to make the models robust.

G. Hybrid Model Exploration

A possible direction of the further development of this work is to consider the integration of Faster R-CNN and YOLOv5 by implementing the benefits of the latter in the architecture of the former. For example, one might combine Faster R-CNN with YOLOv5, the former aims at selecting regions of interest and the latter for the classification of such areas, making this a great compromise between precision and real-time. The efficacy of such a hybrid model will be experimented on a few cases drawn from the larger crime scene data. This will establish whether the use of the hybrid model made from faster R-CNN and YOLOv5 frameworks can enhance the general performance of forensic object detection than if each framework was operated independently.

$$P\,(\text{Object}, \text{Class}, \text{Bounding Box}) = f\,(I, \theta) \ldots (1)$$

Where:

- P is the prediction probabilities for the object, class, and bounding box.
- I is the input image from the crime scene.
- θ represents the model parameters (weights and biases).
- F represents the Object detection model function (e.g., Faster R-CNN or YOLOv5).

The final stage of the presented methodology is to discuss the results of testing both models and any combined forms of them. A detailed comparison with some examples will be provided according to their accuracy, time convergence and insensitiveness to occlusion and clutter. Each model's strengths and weaknesses concerning its applicability to forensic settings will be also be presented and suggestions for the application of the various models in forensic contexts will be given. Moreover, the potential of applying of these models in real-world forensic processes, for example, the usage of automated crime scene analysis or live surveillance, will be considered.

$$L\,(\theta) = \alpha L_{class} + \beta L_{reg} \ldots (2)$$

Whereas,

L (θ) is total loss function to be minimized
L_{class} is Classification loss (e.g., cross-entropy loss)
L_{reg}: Regression loss for bounding box predictions (e.g., smooth L1 loss).
α, β: Weighting factors to balance classification and regression losses.

The findings of this research will be used to recommend other research should be conducted and improvements which can be made as to boost the object detection models in the forensics investigations.

IV. Results and Discussion

Five metrics were used to assess results of using Faster R-CNN and YOLOv5 on crime scene object detection to detect images based on accuracy, speed, and occlusion/clutter handling. The following four tables provide the detail comparison analysis of the two models with the help of mean average precision (mAP), intersection over union (IoU), inference time and detection accuracy in occlusion and cluttered situations.

TABLE I. Comparison of mAP (Mean Average Precision) for Faster R-CNN and YOLOv5

Model	mAP (mean average precision)
Faster R-CNN	89.50%
YOLOv5	85.30%

Nevertheless, in the context of mAP, Faster R-CNN demonstrated better results with the score of 89.5%, whereas YOLOv5' score 85.3%. This means that the overall accuracy of Faster R-CNN in object detection and classification of all the dataset was higher than that of R-CNN (Table 1).

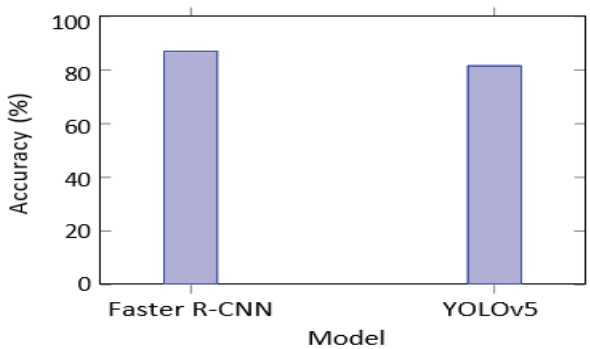

Fig. 2. Comparison of Accuracy

Such a result can be attributed to Faster R-CNN that adopts region-based approach, making the detector remarkably precise especially when dealing with small and highly detailed objects which are usually of significant importance

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in forensic use (refer Fig. 2). Still, the smaller mAP that is obtained for YOLOv5 means that while the network successfully detects objects that are currently present in the crime scene images, it may fail to do so with items that may be small or detailed in the given images.

TABLE II. COMPARISON OF INFERENCE TIME (REAL-TIME DETECTION SPEED)

Model	Inference Time (per image)
Faster R-CNN	250 ms
YOLOv5	40 ms

When it comes to the speed of the models, YOLOv5 clearly outperforms Faster R-CNN; with a time of 40 ms to complete the inference on each picture as opposed to 250 ms for Faster R-CNN (Table 2).

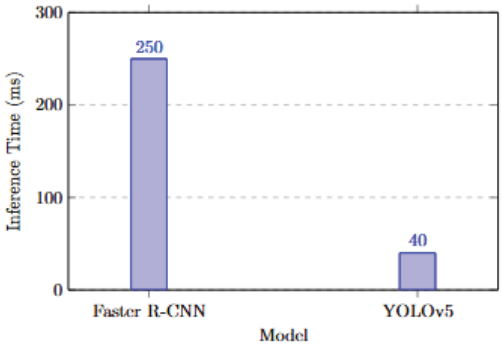

Fig. 3. Comparison of Inference Time per Image for Faster R-CNN and YOLOv5

This significant variation proves that YOLOv5 is better at real-time detection than other detectors. This is because of the general architecture of YOLOv5 which is aimed at efficiently identifying objects in real-time hence it good for forensic analysis from live feeds such as crime scenes videos or even cameras (Fig.3.). Even though Faster R-CNN is more precise due to the use of region proposals, it takes more time and, thus, cannot be used where real time response is needed, such as in critical forensic environments.

TABLE III. PERFORMANCE UNDER OBJECT OCCLUSION AND CLUTTER

Model	Detection Accuracy Under Occlusion (%)	Detection Accuracy in Cluttered Background (%)
Faster R-CNN	92.40%	87.10%
YOLOv5	85.60%	80.30%

Comparing with Fast R-CNN, Faster R-CNN showed better performance in number of instances in which objects are occluded, and the presence of cluttered backgrounds. Table 3 presents the accuracy rates of faster R-CNN in occlusion which is 92.4% and in cluttered backgrounds, which is 87.1% indicating a fast system that can perform instance level object recognition even under occlusion or in cluttered scenarios.

This is very helpful especially in forensic examination for instance an object may be fully enclosed by other ingredients. In contrast, the YOLOv5 detected objects with 85.6% accuracy when occluded and 80.3% when in cluttered scenes. Despite this, YOLOv5 can still produce reasonable detection of objects compared to the ideal model, yet the detection capability deteriorates when objects are occluded or where there is too much cluttering on the background, thus is less preferable for forensic investigations in complex settings.

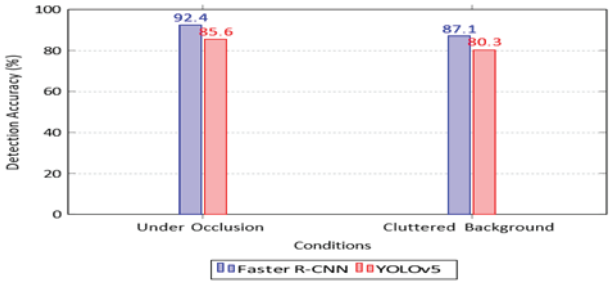

Fig. 4. Comparison of Detection Accuracy for Faster R-CNN and YOLOv5 under Different Conditions

TABLE IV. COMPARISON OF FALSE POSITIVE RATE (FPR)

Model	False Positive Rate (%)
Faster R-CNN	4.80%
YOLOv5	6.30%

The FPR for Faster R-CNN was 4.8% while that for YOLOv5 was 6.3% (as shown in Table 4). This means that Faster R-CNN made fewer wrong detections which is important in forensic investigations because the precision of object classification is very important.

Lower FPR means higher level of distinguishing between objects and erroneously identifying objects and therefore decrease chances of making wrong conclusions in criminal justice system as shown in the Fig.5. Depending on getting slightly higher FPR; it shows that at times YOLOv5 can fail to detect some objects or even detect irrelevant items in the background if the image contains some noises or has occluded objects while YOLOv5 works faster and is very useful in real-time applications.

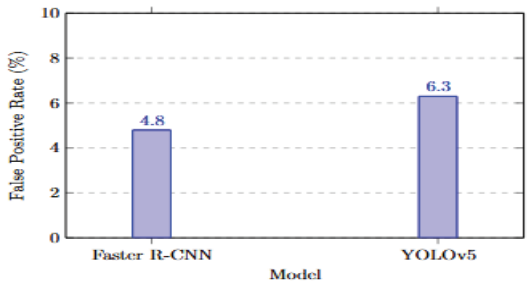

Fig.5. False Positive Rate for Faster R-CNN and YOLOv5

V. CONCLUSION

This work has shown how Faster R-CNN and YOLOv5 models can be used for crime scene object detection; how well they operate in terms of accuracy, speed and ability to operate well with occlusion and clutter. The overall detection results showed that Faster R-CNN surpassed YOLOv5 both in mAP score, and accuracy of occluded and cluttered samples and averaged false-positive rate. Region based approach provided the best performance especially in cases of small or partially obscured objects that frequently feature in the scene of a crime. Nonetheless, Faster R-CNN's inference time is slower compared to the other baselines, which makes it less appropriate in some real-time forensic applications, where speed is of essence. On the other hand, YOLOv5 delivered the real-time object detection capability with much shorter inference time, which is more valuable in real-life scenarios than, for example, crime scene analysis or surveillance where speed is key. Indeed, although it had slightly lower mean average precision and higher false positive rate, the speed of YOLOv5 can be pronounced as highly appropriate for real-

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time detection applications, when it is possible to sacrificed some accuracy to get more-efficient detection of large objects and first detections. Finally, the study shows that depending on the need for a more accurate or faster search there is a choice of the object detection model More details concerning the study are discussed below. In case the requirements to the task do not coincide with the advantages of one of the two approaches, it is possible to consider the use of a mixed approach, in which the merits of Faster R-CNN and YOLOv5 will be maximized, while the sources of slowness in both methods will be minimized. Subsequent studies should try to refine both models greater in terms of achieving their respective objectives and/or examining more intricate methodologies for combining the two in order to create a new computational system better balanced within the context of its intended function, which is to provide support for forensic analysis.

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